

# **RETAIL REVENUE ANALYSIS**

**Data-Driven Insights for Smarter Retail Decisions**

**Project Report**

**Tools Used: Excel | SQL | Python | Power BI**

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# RETAIL REVENUE ANALYSIS

## INTRODUCTION

Retail Revenue Analysis focuses on analysing retail transaction, customer, product, store, and sales data to understand overall business performance. The project uses Excel, SQL, Python, and Power BI to transform raw data into meaningful insights related to revenue and sales performance and customer purchasing behaviour.

### 1. PROBLEM STATEMENT

The main problem is that raw retail data is difficult to understand and analyse manually. Therefore, the project focuses on analysing sales, customers, products, and store data to identify revenue trends, top-performing products, and customer patterns, helping the business make better data-driven decisions.

**Project Goal:** Convert raw retail data into clear, decision-support insights.

### 2. PROJECT OBJECTIVES

| Objective                 | Purpose                                                  |
|---------------------------|----------------------------------------------------------|
| Analyze Revenue           | Identify overall sales and revenue performance           |
| Identify Top Products     | Find best-selling products based on revenue and quantity |
| Study Customer Behavior   | Understand customer purchase patterns and spending       |
| Compare Store Performance | Analyze revenue across different stores/locations        |
| Track Sales Trends        | Identify monthly and yearly revenue trends               |
| Segment Customers         | Classify customers based on their purchasing behavior    |
| Identify Low Stock        | Detect products that need inventory attention            |
| Support Decision Making   | Provide data-driven insights for business growth         |

### 3. TOOLS AND TECHNOLOGIES

| Tool     | Purpose                                  |
|----------|------------------------------------------|
| Excel    | Data cleaning, formatting & preparation  |
| SQL      | Data querying, joins & business analysis |
| Python   | Data analysis & customer segmentation    |
| Power BI | Interactive dashboard & visualization    |

### 4. EXCEL – DATA PREPARATION AND ANALYSIS

The Excel stage prepares the raw retail data for analysis and creates a structured base for the later SQL, Python and Power BI stages.

- Raw Data □ Data Cleaning □ VLOOKUP / XLOOKUP □ Sales & Revenue Calculations □ Pivot Table Analysis □ Duplicate Identification & Conditional Formatting □ Business Insights.

#### *Key Excel Activities*

- Data cleaning and formatting to improve consistency and usability.
- Using VLOOKUP/XLOOKUP to retrieve and combine related information.
- Using Pivot Tables to summarize business information.
- Identifying duplicate values and applying conditional formatting.

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### 5. SQL ANALYSIS

SQL is used to query and analyse the retail database, combining related tables and summarizing information for business questions.

- Database and table preparation.
- Data cleaning and validation.
- Using joins to combine related data.
- Revenue and sales calculations through aggregation.
- Using LIMIT to focus on top results.
- Analysing top products and customer segmentation.

## 6. PYTHON DATA ANALYSIS

Python, using Pandas, is used for data exploration, aggregation and customer segmentation.

- Importing and preparing data using Pandas.
- Exploring the dataset with `describe()`, `info()` and `value_counts()`.
- Performing data analysis and aggregation.
- Creating RFM features: Recency, Frequency and Monetary.
- Using RFM-based analysis for customer segmentation.

Identifying revenue and customer patterns. Generating business insights from the analytical results.

### 1. Customer Segmentation using RFM

RFM analysis provides a structured way to study customer purchasing behaviour. The three features used in the project are Recency, Frequency and Monetary. These features are used to understand how recently customers purchased, how frequently they purchased, and the monetary value associated with their purchases.

*RFM:*

- Recency – measures how recently a customer made a purchase.
- Frequency – measures how often a customer purchased.
- Monetary – represents the customer's monetary value/purchase contribution.

### 2. Power bi dashboard

The Power BI stage converts the analytical results into an interactive dashboard for easier interpretation. The supplied project presentation identifies Power BI as the tool for interactive dashboard and visualization.

*Dashboard Purpose*

- Present analytical results in an interactive visual format.
- Help users understand revenue and sales performance.
- Support customer behaviour and segmentation analysis.
- Make business insights easier to communicate and interpret.

### 3. End-to-End Project Workflow

Raw Retail Data ☐ Excel Data Preparation ☐ SQL Business Analysis ☐ Python Data

Analysis & RFM Segmentation □ Power BI Dashboard □ Business Insights.

This workflow connects data preparation, querying, analytical modelling and visualization into a single retail revenue analysis process.

#### 4. Key Analytical Areas

| Area                        | Analysis Focus                                                    |
|-----------------------------|-------------------------------------------------------------------|
| Revenue & Sales Performance | Understanding overall sales and revenue performance.              |
| Top Products                | Identifying products performing strongly by revenue and quantity. |
| Customer Behaviour          | Studying purchasing patterns and customer spending.               |
| Store Performance           | Comparing revenue across stores or locations.                     |
| Sales Trends                | Tracking monthly and yearly revenue patterns.                     |
| Customer Segmentation       | Classifying customers based on purchasing behaviour.              |

#### 5. Business Value of the Project

- The project transforms raw retail data into a structured analytical workflow.
- It combines spreadsheet analysis, database querying, Python-based analysis and interactive visualization.
- It provides a common analytical view of revenue, products, customers, stores and sales trends.
- It supports data-driven interpretation rather than relying only on manual inspection of raw data.
- It demonstrates how multiple analytics tools can be integrated into one end-to-end project.

#### 6. CONCLUSION

The Retail Revenue Analysis project successfully transformed raw retail data into meaningful business insights using Excel, SQL, Python, and Power BI. The project demonstrates how data analytics can transform complex retail data into clear and actionable business insights, helping the business understand performance and support data-driven decision-making.

## 7. FUTURE SCOPE

- Automate data refresh and dashboard updates.
- Extend customer segmentation with additional behavioural features.
- Add more advanced forecasting and predictive analysis when sufficient historical data is available.
- Introduce additional operational KPIs and inventory monitoring.
- Build more interactive drill-through and reporting views in Power BI.

### Source Basis

This report is based on the uploaded Retail Revenue Analysis project presentation. The report preserves the project scope, tools, objectives, workflow and conclusion supported by that source.

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